OR-AS Operations Research Applications and Solutions	Case Name: Shopping Village Walkways	Sector	**Construction (commercial building)**
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources Schedule with costs
		Risk Analysis	Random simulation One of nine std. scenarios User defined distributions
Submitted by	Mieke De Saedeleer		
Date	2016	Project Control	Automatic tracking Tracking based on user input
File Name	**C2016-07 Shopping Village Walkways**		

1. Project description

Project authenticity

This project involves the construction of walkways within a shopping village, organized into multiple zones and sections. It begins with site accommodation and setup, followed by paving works carried out progressively across different zones. The work is executed in a structured, section-by-section approach to ensure efficient coordination and minimal disruption. Finally, the project concludes with snagging and handover of the completed walkways.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	123
Planned Duration (PD)	224
Budget At Completion (BAC)	930179.09
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	95%
Activity Distribution (AD)	98%
Length of Arcs (LA)	0%
Topological Float (TF)	99%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	1%	9%	10.54
CRI-rho	50%	5%	10.54
CRI-tau	100%	0%	-10.54

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	21%	10%	5.02
SI	31%	11%	4.48
SSI	2%	8%	10.51
CRI-r	8%	11%	5.86
CRI-rho	9%	11%	5.15
CRI-tau	7%	13%	6.09

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	15%	-11%
PV - SPI	31%	20%
PV - SCI	31%	20%
ED - 1	16%	-5%
ED - SPI	31%	20%
ED - SCI	31%	20%
ES - 1	19%	-18%
ES - SPI(t)	17%	8%
ES - SCI(t)	17%	8%

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0%	0%
CPI	0%	0%
SPI	21%	21%
SPI(t)	15%	15%
SCI	21%	21%
SCI(t)	15%	15%
0.8 CPI + 0.2 SPI	11%	11%
0.8 CPI + 0.2 SPI(t)	5%	5%

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 11 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-454.80	-115096.58	-60.90	1.00	0.80	0.69	0.96
std dev	1012.45	88503.77	44.84	0.00	0.20	0.14	0.04
final	-2578.13	0.00	-107.88	1.00	1.00	0.68	1.00

2.3.3.2. Time forecasting

41% LATE

Late/early

PD	224	Real Duration	316

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	179.73	14.87	43%	100%
PV - SPI	216.36	76.98	37%	100%
PV - SCI	216.36	76.98	37%	100%
ED - 1	193.64	21.71	39%	100%
ED - SPI	232.36	70.05	32%	100%
ED - SCI	232.36	70.05	32%	100%
ES - 1	203.64	31.63	36%	100%
ES - SPI(t)	239.45	45.69	24%	100%
ES - SCI(t)	239.45	45.69	24%	100%

2.3.3.3. Cost forecasting

BAC	930179.09	Real Cost	932757.25	Over/under budget	0.2%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	930633.86	1012.45	0%	0%
CPI	930634.63	1014.08	0%	0%
SPI	1142744.16	386772.12	23%	-23%
SPI(t)	1083579.93	220210.06	16%	-16%
SCI	1142744.93	386771.67	23%	-23%
SCI(t)	1083581.28	220209.06	16%	-16%
0.8 CPI + 0.2 SPI	953877.85	36507.59	2%	-2%
0.8 CPI + 0.2 SPI(t)	950710.44	26226.92	2%	-2%